higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T740**(E)**(D2)T

NOVEMBER EXAMINATION

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

(8080205)

2 December 2015 (X-Paper)
09:00–12:00

NATED

This question paper consists of 5 pages and 1 formula sheet of 2 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL INSTRUMENTS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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SECTION A

QUESTION 1

1.1 Define, with the aid of a neat sketch, FOUR types of energies existing in a pipe line. (5)

1.2 With the aid of these energies and the application of Bernoulli's theorem for liquids, prove that $Q = C.A.E \sqrt{2gh}$ for a orifice plate. (12)
[17]

QUESTION 2

2.1 Describe, with the aid of a sketch, the principle of operation of a swirl meter. (10)

2.2 Show, by means of a sketch with annotations, how pressure and temperature are compensated for when using swirl meters. (4)
[14]

TOTAL SECTION A: 31

SECTION B

QUESTION 3

3.1 Sketch and describe the operation of a target flow meter. (6)

3.2 Name TWO advantages and TWO disadvantages of the target flow meter. (4)
[10]

QUESTION 4

Make a labelled sketch of a falling-ball viscometer. **[4]**

QUESTION 5

Sketch and describe the operation of a photo-electrical hydrometer. **[10]**

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QUESTION 6

6.1 What is understood by the isopotential point in pH measurement? (2)

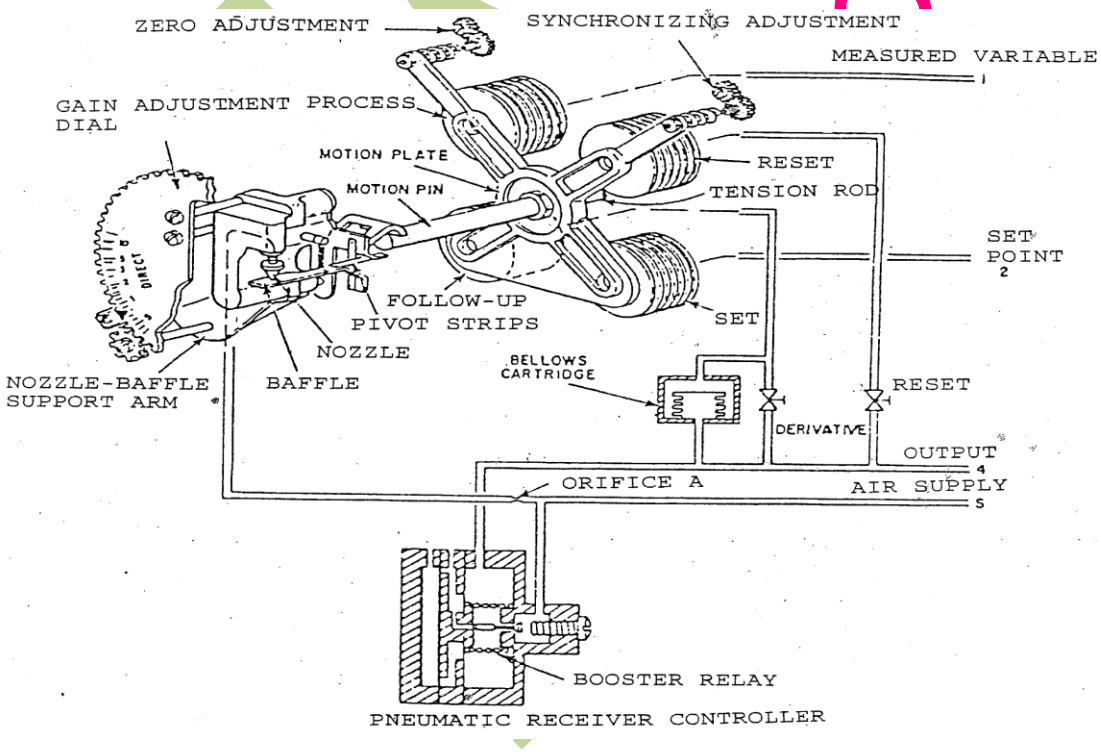
6.2 What does it mean if the pH meter registers a 9 on it scale? (3)

6.3 Name FOUR requirements that the ideal housing material for electrodes must comply with. (4)
[9]

TOTAL SECTION B: 33

SECTION C

QUESTION 7



Describe in detail, with the aid of the above sketch of a pneumatic receiver controller the following:

7.1 The conditions when the controller is in balance. (3)

7.2 The conditions if the measured variable increases by 10 kPa.
The gain adjustment is 1.0 and no reset (integral) or derivative (rate) action is generated. (4)

7.3 What is the proportional band of the controller if the setting is as indicated in QUESTION 7.2? (1)

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7.4 How would you reverse the action on this type of controller? (1)

7.5 Explain what is meant, when the intergral action of a controller is marked in 'R.P.M' or 'M.P.R.' (1)

7.6 Explain the term *derivative action*. (2)

7.7 What is meant by interaction with regard to most controllers? (2)

7.8 Give THREE advantages of a live signal and ONE advantage of a zero-based signal as used in electronic controllers. (4)
[18]

QUESTION 8

8.1 Sketch and describe the operation of a basic proportional electronic controller which makes use of an operational amplifier. (8)

8.2 Sketch a block diagram of a cascade loop. (5)

8.3 Explain what is meant by closed-loop control. Use a diagram to motivate your answer. (5)
[18]

TOTAL SECTION C: 36
GRAND TOTAL: 100

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INDUSTRIAL INSTRUMENTS N5

FORMULA SHEET

$W = 359{\cdot}2\, C\, Z \epsilon E d^2 \sqrt{(h\rho)} \quad \rho \quad R_d = W/158\, \mu d$

$Q = 359{\cdot}2\, C\, Z \epsilon E d^2 \sqrt{(h\rho)} \quad \rho \quad R_d = Q\rho/158\, \mu d$

$Q_g = 2238\, CZEd^2 \sqrt{(h/\rho)} \quad R_d = Q_g \rho/98{\cdot}6\, \mu d$

$m = (d/D)^2 \qquad E = 1/\sqrt{(1-m^2)}$

$W = 001252\, CZ\epsilon E d^2 \sqrt{(h\rho)} \qquad R_d = 3{\cdot}54\, W/\mu d$

$Q = 001252\, CZ\epsilon E d^2 \sqrt{(h/\rho)} \qquad R_d = 354\, Qd/\mu d$

$m = (d/D)^2 \qquad E = 1/\sqrt{(1-m^2)}$

$$N = \frac{W}{0{\cdot}01252\, D^2 \sqrt{(h\rho)}} = \frac{Q\sqrt{(\rho)}}{0{\cdot}01252\, D^2 \sqrt{(h)}}$$

$mE = N/CZ\epsilon$

$CmE = N/Z\epsilon \qquad mE = CmE/C$

$$R_d = \frac{3{\cdot}54W}{\mu D\sqrt{(m)}} = \frac{3{\cdot}54Q\rho}{\mu D\sqrt{(m)}}$$

$d/D = [(mE)^2/1 + (mE)^2]^{\frac{1}{4}}$

$W = 1{\cdot}252\, Ud^2 \sqrt{(\rho P)}$ *vir kritieke vloei/ for critical flow*

$d = [W/1{\cdot}252\, U\sqrt{(\rho P)}]^{\frac{1}{2}}$ *vir kritieke vloei/ for critical flow*

$$N = \frac{W}{359{\cdot}2\, D^2\sqrt{(h)}} = \frac{Q\sqrt{(\rho)}}{359{\cdot}2\, D^2\sqrt{(h)}} = \frac{Q_g\sqrt{(\rho)}}{2238\, D^2\sqrt{(h)}}$$

$mE = N/CZ\epsilon$

$CmE = N/Z\epsilon \qquad mE = CmE/C$

$$R_d = \frac{W}{158\mu D\sqrt{(m)}} = \frac{Q\rho}{158\,\mu D\sqrt{(m)}} = \frac{Q_g\rho}{986\mu D\sqrt{(m)}}$$

$d/D = [(mE)^2/1 + (mE)^2]^{\frac{1}{4}}$

$W = 1890\, Ud^2\sqrt{(\rho P)}$ *vir kritieke vloei/for critical flow*

$d = [W/1890U\sqrt{(\rho P)}]^{\frac{1}{2}}$ *vir kritieke vloei/for critical flow*

1 kPa = 102 mmWD = 102 mmWC

1 lb/ft^3 = 16,0183 kg/m^3

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Atmosferiese druk/Atmospheric pressure = 101,325 kPa

Gravitasieversnelling/Gravitation acceleration = 9,81 m/s²

For/ Vir D + $\frac{D}{2}$ *-tappe/tappings en/and flenstappe/flange tappings* $\frac{h}{Pa} \times 27{,}2 = \frac{kPa}{kPa} \times 27{,}2$

$$Q = \frac{8}{15} \quad Tan \frac{\theta}{2} \sqrt{2g.H^5}$$

$$Q = \frac{2}{3} \quad B \sqrt{2g.H^3}$$

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